

"PREDICTING OPERATIONAL EFFICIENCY: A RANDOM FOREST APPROACH TO NYC TAXI TRIP DURATION"

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ABSTRACT. Operational uncertainty regarding trip duration and passenger acquisition represents a significant inefficiency within the New York City taxi network. We address these challenges by leveraging a random forest regression model against the extensive historical dataset provided by the NYC Taxi and Limousine Commission to predict route duration and aid in next-location strategies. By analyzing millions of meter-reported trip records, we employ data cleaning and machine learning techniques to derive trip time estimates and pickup location recommendations.

1. RESEARCH QUESTION AND OVERVIEW OF THE DATA

New York City is the most densely populated city in the United States. As such, passengers and operators of their taxi network are subject to levels of uncertainty regarding route duration and the optimal location for passenger acquisition. Using a compilation of years of trip record data provided by the New York City Taxi and Limousine Commission (TLC), we aim to leverage data analysis and machine learning techniques to develop a model that will forecast answers to these complementary questions. If successful, this model would be a valuable tool in a vital sector of New York City's transportation network.

The TLC Yellow Taxi trip record dataset documents millions of individual trips dating as far back as January of 2009. [1] It is compiled from driver usage of taxi meters, exposing it to a limited degree of human error. It includes as many as 20 data features for each entry, including date and time of passenger pickup and drop off, taxi zone location ID of passenger pickup and drop off, trip distance, and payment information. Altogether, this data encodes traffic patterns throughout the city and zone specific taxi demand with precise time resolution. This makes it an ideal dataset for pursuing questions of travel time and passenger pickup recommendations.

We expect that our results will align with known traffic patterns across New York City. We also expect our results regarding trip time and known taxi demand will reveal that taxi drivers who seek clients from adjacent zones of higher demand will have an advantage over drivers who wait in their drop

off zones for their next clients. We additionally expect to learn about the most effective time ranges to train a model on for optimal performance.

2. DATA CLEANING / FEATURE ENGINEERING

Despite its virtues, the TLC dataset is subject to numerous undesired artifacts of its method of collection, including manual entry errors and system logging glitches. Hence, to prepare the dataset for predictive modeling, we implemented a pipeline designed to clean and isolate data relevant to our research questions. This involved filtering out of scope records, imputing missing values, and removing outliers.

2.1. Scope Definition and Record Filtering. To ensure our model learns the dynamics of standard transit strictly within New York City, we first applied exclusion criteria based on location and rate type data features. We removed records where the Pickup or Drop off zones were labeled as "Unknown" (264) or "Outside of city" (265), as these IDs lack geographical grounding and extend outside the scope of our model.

Concurrently, we restricted the dataset to trips labeled as Standard Rate (1). We excluded other rate codes, such as Newark (3), Nassau/Westchester (4), and Negotiated Fares (5), because they represent transit outside the city as well as traffic patterns that deviate significantly from standard city driving. By utilizing standard metered trips, we hope to improve the model's ability to generalize typical traffic conditions within the city, although this choice necessarily limits the model's applicability.

Finally, after accounting for duplicate data and outliers, as well as engineering additional features, we chose to limit the training of our model strictly to information that would be known to the driver at the moment a trip was started. Hence, after row cleaning and feature engineering, all columns were dropped except those corresponding pickup datetime and location and drop-off location. We considered including trip distance, as we supposed this would be a valuable predictor in trip time, but concluded that including trip distance could be "cheating," as this datum may not be exactly known to a driver at the moment of departure and could be variable depending on route.

2.2. Missing Data. Analysis revealed missing values in the Rate Code ID, Passenger Count, and Congestion Surcharge columns. We considered data imputation for missing Rate Code ID's, but given the quantity of data in our possession and the uncertainty of such entries' conformity to our desired scope, we opted to discard these entries. Furthermore, after evaluating background data on the two features, we concluded that missing data in the Passenger Count and Congestion Surcharge columns did not compromise overall trip data of interest.

2.3. Duplicate Data. This is an interesting aspect of the data. A cursory inspection revealed no exact duplicate entries, however there are thousands of duplicates in a given month which have equal values for every attribute except payment amounts, which are negated between the two. This likely represents a voided transaction of some kind. However, it is reasonable to assume that even though the financial values may be unusual, the physical values describing the geography and timing of these trips are legitimate. That is to say that the described taxi ride for a given entry actually happened, and so its data can be used to train a model. Hence, it is sufficient to just filter out the duplicates with negative payment values, which we accomplished as we filtered away outliers.

2.4. Outliers. We evaluated payment, distance, and trip time rows for outliers. On the low end, we eliminated entries with negative or 0-valued entries in each of these categories. On the high end, we cut values that exceeded the 98th percentile in their respective categories. We hypothesized that such large values in these categories could result from technical errors or from a driver forgetting to reset his or her meter.

2.5. Datetime Transformations. We represented the pickup datetime data with some trigonometric transformations, as suggested in the assignment instructions. That way, our model can understand, for example, that December is close to January and 11:59 p.m. is close to midnight. In this preprocessing, we represent one 'circle' for day of the year and another for time of day using the formulas $\sin(\frac{2\pi t}{T})$ and $\cos(\frac{2\pi t}{T})$, where t represents the time entry of interest (time of day or day of year) and T represents the period of interest (24 hours or 365.25 days). We also considered that day of the week could have an impact on traffic conditions, and might be represented by either the same cyclical transformations or by a one-hot encoding, but in this model any such attributes are omitted.

2.6. Target Encoding. The foremost hurdle in building our random forest regressor was handling the pickup and drop-off location ID values. The map of NYC is divided into 263 geographically designated taxi zones, which are represented in the data by a corresponding integer. One-hot encodings of these values for model training are computationally infeasible given our memory restrictions. We considered reducing the zones of interest to maintain one-hot encodings, but this dramatically decreases the applicability of our model. Instead, we here applied a method known as target encoding. This method substitutes in an average of trip times corresponding to pickup and drop-off zones to handle high-cardinality features [2].

3. DATA VISUALIZATION AND BASIC ANALYSIS

A basic analysis of the TLC dataset allowed us to verify its properties before constructing a more detailed model. Specifically, we were able to analyze the following:

- The range of data for trip distance and trip duration
- The frequency of trips by distance

Doing this preliminary analysis was valuable as it allowed us to gain a fundamental understanding of this dataset before applying machine learning techniques to it.

3.1. Ranges of Data Categories. We expected that shorter trips, both in time and distance, would be far more frequent than longer ones. However, we also expected multiple outliers from longer trips to be present. In Figure 1, we see that this behavior is confirmed. While there are trips even as far up as 90, the vast majority of data points fall within the first 10 values.

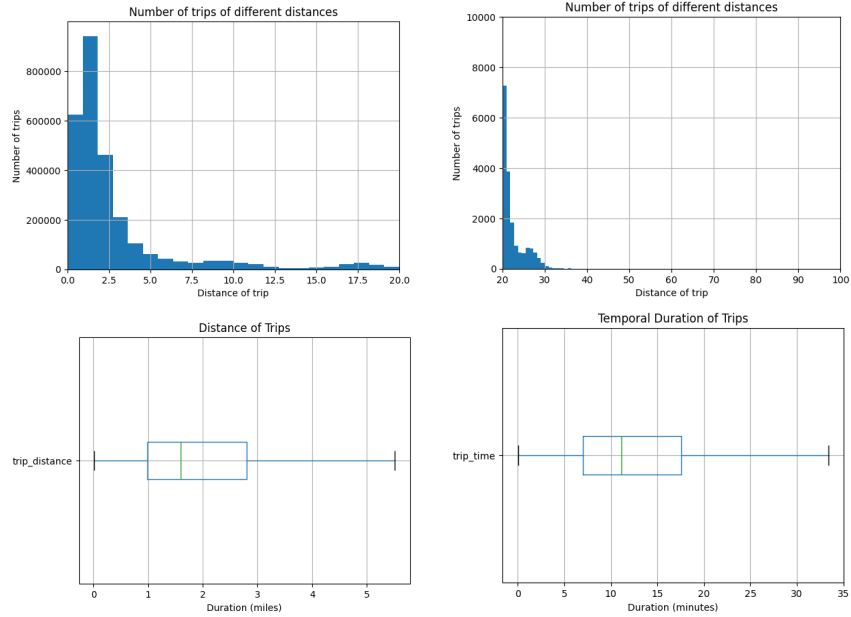


FIGURE 1. The two histograms show that the majority of the trips within the data set were of duration 20 or less, but that longer trips still existed. The two box charts show the range of non-outlier values of the dataset.

4. LEARNING ALGORITHMS AND IN-DEPTH ANALYSIS

We employ an ensemble learning approach. Specifically, we utilize the Random Forest adapted as a regressor for trip time prediction.

Due to significant memory restrictions, we trained the Random Forest on samples of 1.2 million entries of processed data over the course of 12 months, rather than the 30+ million entries that those 12 months of processed data represent. Based on performance results of a Grid Search method for hyperparameter selection, we built our random forest so each tree has a maximum

depth of 30, a minimum number of 25 samples in each leaf, to test on all possible features, and due to memory restrictions our forest is composed of 100 estimators. Given the nature of our data, we were particularly surprised that our forest performed best without attribute bagging.

The test results of this random forest revealed an average absolute error of 3.943 minutes, a mean squared error of 33.613, and an r^2 correlation of 0.678. This means that given a pickup and drop-off location, the model can predict a trip's duration to within about 4 minutes accuracy. Given that the majority of trips in the data set are under 35 minutes long, a 4 minute average error is significant but still useful.

5. ETHICAL IMPLICATIONS AND CONCLUSIONS

The ethical implication of this work is an important consideration to make anytime machine learning is used. In the case of the work of this model, we expect the primary impact caused to be influencing where taxi drivers frequent when not driving a passenger to their destination. Generally, we expect this to guide taxi drivers to higher frequency areas where they are more likely to quickly find another person in need of a taxi. This should have a positive impact on both company earnings and also on the experience of individuals seeking service from a taxi- the less time a taxi spends between passengers, the more people have been served by that taxi driver.

However, this model is not without risk. It is possible that low-density areas could be negatively impacted by the adaptation of this model. If an area is less busy than the surrounding neighborhoods, it is likely taxi drivers would be guided to avoid this area. Further, this could lead to feedback loops that self-reinforce this deficiency, as anywhere that taxi drivers are lead to avoid will have less data and appear less productive as a result.

CODE AVAILABILITY

The data cleaning, feature engineering, and model training code used for this analysis is available at the following GitHub repository:

<https://github.com/tpace31/Volume-3-Project>

REFERENCES

- [1] NYC Taxi and Limousine Commission, “TLC Trip Record Data,” City of New York, Nov. 2024–Oct. 2025. [Online]. Available: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>.
- [2] D. Micci-Barreca, “A preprocessing scheme for high-cardinality categorical attributes in classification and prediction problems,” *SIGKDD Explor. Newsl.*, vol. 3, no. 1, pp. 27–32, Jul. 2001.